

Safety Factor — play-through review

Theme `midway` · AP Physics 1, algebra-based, all eight units (grade 12) · 15 days, ~46 stops · reviewed 2026-08-21 by reading the full book (`books/midway.yml`), solving every question and every DERIVE branch, and reading opening, stakes, warm-ups and ending in play order.

Verdict

The best-taught physics in the catalogue, and the clearest case in the repo for what a *place* can do to a syllabus. Seven rides, one per unit: the drop tower is kinematics, the coaster is energy, the wheel is torque, the flume is fluids. The player does not walk to a topic — they walk to a machine, and the machine happens to be the topic. Nothing else here is that clean.

Its argument is the best-balanced in the set. Marsh holds that forty-one years of safe running is evidence; Vey holds that evidence of having got away with it is not a margin. **Marsh is right on day 4** — Brennan's carousel settings turn out to be correct to within a degree of the derivation. **Vey is right on day 9** — the loop was regraded in 1998 and the drawing was never updated, so the crown demands 8.5 m/s where the 1974 paper implies 7.4, and the certificate's margin of 1.38 has actually been 1.20 for twenty-five years. Both are vindicated by arithmetic the player performs, and neither is made a fool of.

The DERIVE stops are the campaign, and they are the best-authored panels in the repo. Every wrong branch is a real student mistake that stays algebraically valid: $v = gr$ (dimensionally wrong, caught by the number coming out at 73), $v = \sqrt{(g/r)}$ (a bigger loop needing less speed), $N = mg$ at the crown (a comfort criterion mistaken for a limit, 40% above the true minimum), $F = m\Delta v\Delta t$ (the division inverted, giving 15.9 N where the answer is 490). The one that impressed me most survives deliberately: "8.5 *against* 7.4 drawn, a difference of 1.1 m/s in the margin" — the subtraction is correct and the framing is the defect, because quoted as a difference it reads like small change where the ratio is 15%.

Answerable: 46/46. Everything settles from the card. **Sense:** Excellent within a day. Three cross-day inconsistencies about the pirate ship's drive (MW-01) and one about the wheel (MW-04). **Level:** Right for AP Physics 1 and unusually generous with the algebra. Ten DERIVE stops is the most in the set and it is the right choice for this fiction. **Fun:** The highest here. Certifying a shut amusement park with a hostile inspector and an owner counting advance bookings is a better frame than any disaster in the catalogue, and the props layer visibly opens the park as the campaign advances.

The questions, solved

All arithmetic reproduces. Verified: $\sqrt{(2 \times 9.8 \times 42)} = 28.7$ m/s at the foot of the tower; $28.7^2 \div (2 \times 9.0) = 45.8$ m/s² = 4.67 g in the fins; $6 \times \cos 30^\circ = 5.2$ m as the torque arm; $2\pi \times 6.2 \times 8 \div 60 = 5.2$ m/s and $5.2^2 \div 6.2 = 4.35$ m/s² at the outer horse, so $F = 78 \times 4.35 = 340$ N on the pole; $320 \times 3.1 \div 610 = 1.63$ m/s off the bumper collision; $2\pi\sqrt{(8.6/9.8)} = 5.89$ s for the ship and $4\pi^2 \times 8.6 \div 5.90^2 = 9.75$ m/s² for g; $\sqrt{(2 \times 9.8 \times 8.6 \times (1 - \cos 48^\circ))} = 7.5$ m/s through the bottom of the swing and $1 + 7.5^2 \div (8.6 \times 9.8) = 1.67$ seat-g; $\sqrt{(9.8 \times 7.4)} = 8.52$ m/s

at the measured crown against $\sqrt{(9.8 \times 5.6)} = 7.41$ drawn, a ratio of 1.150, and $1.38 \div 1.150 = 1.20$ — the margin the whole campaign turns on, and it lands exactly.

Two pieces of physics I want to record as unusually good. The ship's large-amplitude correction is **right and never stated as a formula**: 5.90 s at low angle, and the 48° operating swing measured at 6.15 s, which is $5.90 \times (1 + \theta^2/16)$ to three figures. The campaign refuses to let the player assume the small-angle result outside its range and makes them go and measure — that is the correct lesson and almost nothing else in the catalogue does it. And the day-12 VERIFY reconciles 4.7 g average with a 5.4 g peak by naming the mechanism: a magnetic brake makes force in proportion to speed, so it bites hardest at entry.

Implemented since this review

- **MW-01, the ship's drive interval.** One value: 5.85 s, which makes day 12's "three tenths of a second" exact. The ending now says the drive was retimed **from 5.85 to the 6.15 the ride actually swings at**, and adds why — the small-angle formula gives 5.90 and this ride does not run at a small angle.
- **MW-02, the drop-test VERIFY.** `passRatio` 0.85 → 0.93, so the floor is 5.02 g and day 2's 4.7 g average no longer passes the stop that exists to rule it out.
- **MW-03, the EVADE card.** Retitled *Delia Marsh wants an opening date today*, with the job attached and the pronoun corrected.
- **MW-05, both halves.** A new day-1 BALLPARK — 640 kg of ballast at $45.8 \text{ m/s}^2 = 29.3 \text{ kN}$ — computes $\Sigma F = ma$, which **cleared five inversions from** `equation-debt.json` **in one stop**. And the day-9 margin CHOICE became a BALLPARK computing $10.2 \div 8.5 = 1.20$ against $10.2 \div 7.4 = 1.38$, so `safety factor = capacity ÷ demand` is computed. **Midway is now 12/12 equations computed with no rows in** `curriculum-debt.json`, and the conversion invariant confirms nothing was lost.
- **MW-07** numeral damage (18 occurrences, including two inside DERIVE branch notes) and **MW-08** the three place strings.

Everything else in the findings table below is unchanged, and `npm run check` is green on this theme.

Findings

ID	Sev	Where	Issue	Proposed fix
MW-01	FIX	The pirate ship's drive interval — days 9, 12 and the ending	Three numbers for one setting, and the ending contradicts the campaign's own lesson. Day 9's DIAGNOSIS lists the drive as 5.60 s (fixed since 1996) . Day 12's CHOICE scene says the drive " pushes every 5.85 seconds ", and its question — "the drive and the operating swing disagree by three tenths of a second" — only works from 5.85 ($6.15 - 5.85 = 0.30$; from 5.60 it is 0.55). Then the ending says the ship "opened after its drive	Pick one drive interval and propagate it. 5.85 is the one to keep: it makes day 12's "three tenths" exact and day 9's DIAGNOSIS reads the same either way. Then the ending must say retimed from 5.85 to the 6.15 the ride actually swings at , with the point kept: the small-angle formula gives 5.90 and the ride does not run at a small angle. That

was retimed from **5.60 seconds to the 5.90** the pendulum actually keeps." But 5.90 is the *low-angle* period, and both stops' keyed answer is to retime to the **measured operating** period of 6.15 — which is the one thing this campaign spends two stops establishing may not be assumed from the formula. So the closing card reverses the lesson on the way out.

single sentence change turns the ending from a contradiction into the campaign's best summary of itself.

MW- 02	FIX	Day 12 VERIFY, "What the drop test has to show"	The acceptance band admits the answer the stop exists to rule out. <code>truth: 5.4</code> , <code>passRatio: [0.85, 1.15]</code> → anything from 4.59 to 6.21 g passes. Day 2's average deceleration is 4.7 g , and the whole subject of this stop is that the peak is <i>above</i> the average because a magnetic brake bites hardest at entry. A player who predicts 4.7 — i.e. who has not understood the stop at all — passes at 0.87 of truth.	Tighten the low side. <code>passRatio: [0.93, 1.15]</code> puts the floor at 5.02 g, above the average and still forgiving of anyone who reasons "the peak must be somewhat higher" without pinning it. The asymmetry is correct here and worth keeping: over-predicting the peak is the conservative error.
MW- 03	FIX	<code>warmups.evade</code>	The quarry is the park's owner, and the card calls him " He ". The owner is Delia Marsh , referred to as "her" everywhere else in the book and on the opening card. The card also never names her — it says only "the owner who wants an opening date today", so the one warm-up whose whole job is to make you avoid a specific person identifies them by role and then gets the pronoun wrong.	"Delia Marsh, who owns the park, has been on the midway since eight wanting a date for the press...". Fixes the pronoun and satisfies <code>introRule</code> in the same clause.
MW- 04	WORTH	The wheel's arm-nine indication — day 8 vs the ending	Day 8's stake gives the indication as 41 millimetres ; so does the ending. The bolt-group load comes out at 52.7 kN in both. Consistent — but the <i>campaign</i> never says what a fracture assessment would need, and the ending's "whether a 41-millimetre indication can carry it is a fracture assessment nobody on site could do in three weeks" is the only place the player learns that the wheel's fate was never theirs to decide. Day 8 reads as though the load computation settles it.	One clause in day 8's stake or the arm-nine DERIVE's <code>takeaway</code> saying that the load is <i>half</i> the assessment and the other half is a material question outside the fortnight. It makes the wheel staying shut a decision the player understands rather than an outcome sprung on them at the end.
MW- 05	WORTH	<code>curriculum-debt.json</code> , <code>equation-debt.json</code>	$\Sigma F = ma$ is shown on cards from day 1 and computed by no question in the game. Six of the seven rows in Midway's equation-debt file are inversions rooted in it: the energy	Two additions, neither touching a DERIVE (the derivations are the course). (a) A BALLPARK on day 1 or 2 at the tower: the 640 kg ballast

books, centripetal force, torque, the pendulum and fluid pressure are all derived from a base the campaign never computes. This is CLAUDE.md's own worked example and it is still open. Second row: **safety factor = capacity ÷ demand** is uncomputed, because day 9's margin stop is a CHOICE and CHOICE has no relationship, template or worked solution.

carriage in the fins at 45.8 m/s^2 , so $\Sigma F = 640 \times 45.8 = 29.3 \text{ kN}$ through the brake fins — the number is already in the day-12 VERIFY's intervention note, so nothing new has to be invented. That single stop clears six inversions. **(b)** Convert day 9's "What a safety factor says" CHOICE to a BALLPARK computing $8.52 \div 7.41 = 1.150$ and then $1.38 \div 1.150 = 1.20$, which is arithmetic the player currently reads rather than performs, and it is the campaign's headline number.

MW- 06	WORTH Missions 5, 8, 11 and 13	Four days author 4 stops against the loop's 3. All the extra stops are good — the drop-test SEQUENCE, the hydraulic-ram DERIVE, the LOB at the flume, the overnight-levelling SEQUENCE.	Prefer a move to a delete. Day 3 ("Where the energy went") carries three and would host the hydraulic ram; day 14 is the natural home for the overnight levelling, which is a night-before job.
MW- 07	WORTH ~30 stakes, scenes and DERIVE branch notes	Numeral normalisation, and Midway is among the worse-affected because <i>one</i> does a lot of grammatical work in engineering prose: "not 1 line of working anywhere", "the radius is not the 1 on the 1974 drawing", "the expression still looks like the 1 in every textbook", "a comfort criterion being mistaken for a safety 1", "so the 2 forces balance at the crown", "11 mm low at 1 point", "41 years of settings in 1 hand", "100 collisions an hour, and every 1 of them is the same equation". The genuine counts (42 metres, 24 gondolas, 320 kg) are all intact.	Editorial pass. Cross-campaign \$1. Two of these are inside DERIVE candidate why strings, which is the worst place for it — that text is read immediately after a wrong answer.
MW- 08	TASTE Three stops' place strings	"Drop Tower", "Bumper Floor" and "Water Flume" against the areas' own names "The Drop Tower", "The Bumper Cars", "The Log Flume". The player reads two names for one place.	Match the group name. Three lines.
MW- 09	TASTE Day 13 "Three things and one crew"	The sixth campaign to open a late day on "which of the three cannot wait". Midway's DELEGATE version is well adapted — a crew of ride staff, and command keeping the thing that cannot be handed over — but the shape is now a house tic.	Cross-campaign \$6.

MW- TASTE 10	~26 stops	Repeated format boilerplate in background . Sharper here than anywhere: Midway's stop-specific backgrounds are the best in the repo ("why the crown is the critical point", "what the minimum condition means", "why two radii matter so much", "why the prediction is locked"), and on the DERIVE stops they sit <i>after</i> two paragraphs the player has read on every previous DERIVE.	Keep each essay on first use per format. Cross-campaign \$2. This is the campaign that loses most to it.
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Day-by-day notes (short)

- **Day 1** — Forty-one years of notebooks and no working. The tower DERIVE gets 28.7 m/s two ways, the energy PROTOCOL reads a circuit as a ledger, and the ship CHOICE asks why sixty riders do not change a period. A model opening day: the campaign's whole thesis is stated as three questions.
- **Day 2** — What the brake has to take out. 45.8 m/s², nine metres of water, the torque arm people get wrong (the *horizontal* distance, 5.2 m, not the 6 m arm), and the HOLD.
- **Days 3–4** — Where the energy went, and Brennan vindicated. The swing-angle DERIVE arriving at an angle that depends on nothing but speed and radius — not mass, not chain length — and then agreeing with a setting made by hand in 1985, is the best single beat in the campaign.
- **Days 5–7** — The pole, the ram, the collision. The momentum DERIVE and the impulse DERIVE are correctly separated: one is what is conserved, the other is what a neck feels.
- **Day 8** — The period that ignores the load, and g measured with a boat. The cleanest use of "this local value belongs to the site, not the ride, so the tower may use it" anywhere here.
- **Day 9** — The arm and the crack, and the drive that pulls unevenly. MW-01 and MW-04 both start here.
- **Day 10** — The loop. Best day in the campaign, and the one that justifies ten DERIVE stops.
- **Days 11–13** — The impulse on a neck, the margin that moves, the wind rules, and the drop test.
- **Days 14–15** — What goes on the certificate, and what opens. The CASEBOOK sorting seven rides by what can actually be claimed is the right last question for a campaign about signatures.

Opening and closing

Opening: "Corbin Park is an amusement park, and in October the county took away its certificate to open... The signature on all seven is yours... what you inherited is eleven notebooks of settings and timings, forty-one years of them, with not one line of working anywhere. Delia Marsh, who owns the park, says eleven million people have ridden on those numbers unhurt. Marcus Vey, who inspects for the county, calls that a record of having got away with it." Situation, authority as signature, and both sides of the argument, in one paragraph. The best opening card in the set.

Closing: three paragraphs, and the second is the model for what this document keeps asking other campaigns for — the wheel stayed shut, twenty-four gondolas stood still for a season, and Marsh borrowed against next

year to cover it. The third is addressed to the player and ends "A summer of people rode safely and none of them will ever know your name. You will." Keep every word except the ship's retiming figure (MW-01).

Warm-ups

All seven authored and all seven specific to a shut park. FOLLOW — the coaster walk-down at the pace it is actually done, where "the stops are the inspection and the walking between them is just walking" — is the best FOLLOW justification in the repo. CANVASS asks the old crew what the ship used to do at the top of the swing, which plants day 9's fault three days before it arrives. One defect, MW-03.

What to keep

- The rides as the syllabus. Seven areas, eight units, and the player walks between them.
- Every DERIVE distractor that stays algebraically valid, and every why that says what number it produces.
- Marsh right on day 4 and Vey right on day 9, both settled by arithmetic.
- The ship's operating period. A formula the campaign teaches, then declines to let you use outside its range.
- $1.38 \div 1.150 = 1.20$. A certificate that has been wrong for twenty-five years, found by taping a loop.